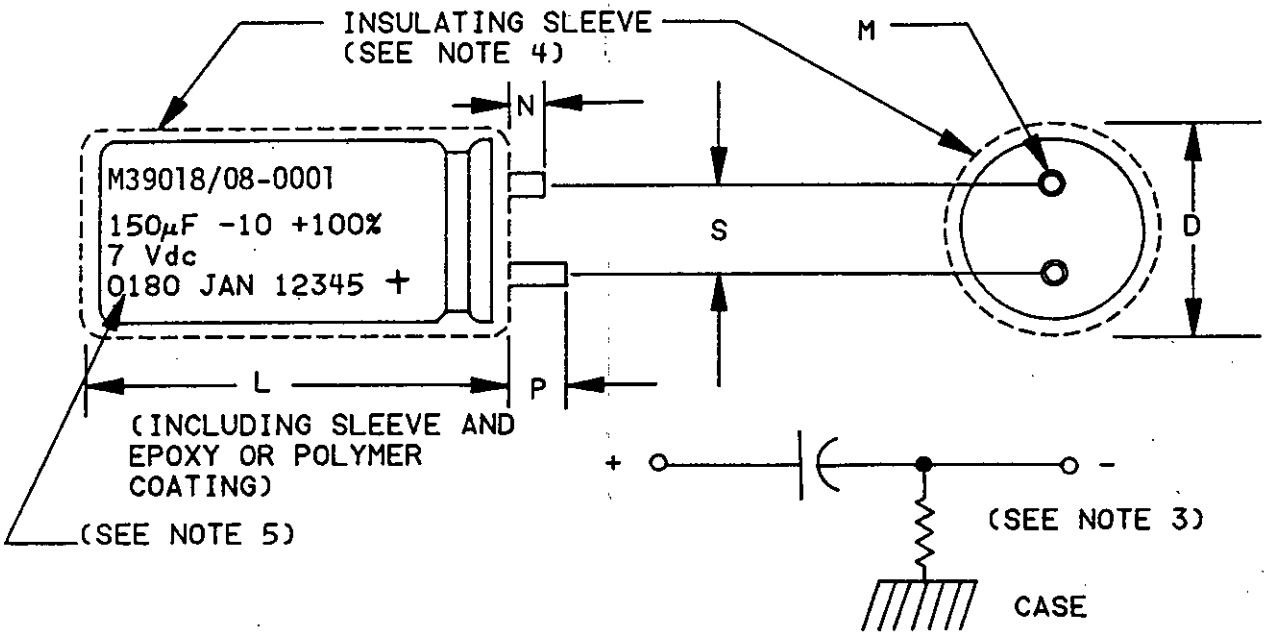
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (ALUMINUM OXIDE), (POLARIZED), NON-ESTABLISHED RELIABILITY AND ESTABLISHED RELIABILITY, STYLES CU01 (INSULATED), AND CUR01 (INSULATED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-39018.



CIRCUIT DIAGRAM

FIGURE 1. Style CU01 and CUR01 insulated capacitors.

FSC 5910

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CASE SIZE	INSULATED DIMENSIONS					
	L±.050(1.27)	D±.015(0.38)	S±.015(.38)	M±.003(.08)	P±.062(1.63)	N±.062(1.63)
A1	0.582(14.78)	0.398(10.11)	0.200 (5.08)	.025 (0.64)	.375 (9.52)	.281 (7.14)
A2	0.690(17.53)	0.398(10.11)	0.200 (5.08)	.025 (0.64)	.375 (9.52)	.281 (7.14)
A3	0.877(22.28)	0.398(10.11)	0.200 (5.08)	.025 (0.64)	.375 (9.52)	.281 (7.14)
B1	1.115(28.32)	0.500(12.70)	0.200 (5.08)	.032 (0.81)	.375 (9.52)	.281 (7.14)
B2	1.379(35.03)	0.500(12.70)	0.200 (5.08)	.032 (0.81)	.375 (9.52)	.281 (7.14)
C1	1.070(27.18)	0.635(16.13)	0.300 (7.62)	.032 (0.81)	.375 (9.52)	.281 (7.14)
C2	1.366(34.70)	0.635(16.13)	0.300 (7.62)	.032 (0.81)	.375 (9.52)	.281 (7.14)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. There is an indeterminate resistance between the metal case and the negative terminal.
4. Insulating sleeve shall overlap the ends of the capacitor body.
5. Example of marking and its relative location on the capacitor body.

FIGURE 1. Style CU01 and CUR01 insulated capacitors - Continued.

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TABLE I. Style CU01 capacitors.

DC rated voltage at 85°C & 105°C	Nominal Capacitance	DC surge voltage at 85°C & 105°C	Equivalent series resistance at 25°C (max)		Maximum impedance at 25°C 100 kHz	Maximum impedance at -55°C (100 kHz)	Maximum DC leakage		Maximum AC ripple current at 85°C (20-100 kHz)	Case size	DASH NUMBER
			120 Hz	10 kHz			25°C	105°C			
Volts	µF	Volts	Ohms		Ohms	Ohms	µA		Amperes		
7	150	10	1.30	0.72	0.60	9.00	2.9	17.4	0.45	A2	0001
10	100	15	1.70	1.00	0.70	10.50	3.3	19.8	0.36	A1	0002
10	220	15	0.79	0.42	0.33	4.95	4.9	29.4	0.63	A3	0003
15	68	20	2.50	1.12	0.85	12.75	4.1	24.6	0.36	A1	0004
15	100	20	1.70	0.70	0.50	7.50	5.0	30.0	0.45	A2	0005
15	150	20	1.20	0.40	0.35	5.25	6.1	36.6	0.63	A3	0006
30	33	40	3.20	2.10	1.50	15.00	5.7	34.2	0.23	A1	0007
30	47	40	2.30	1.35	1.00	10.00	6.9	41.4	0.32	A2	0008
30	100	40	1.30	0.45	0.35	3.50	10.0	60.0	0.63	A3	0009
50	15	75	7.10	3.00	2.20	22.00	6.5	38.7	0.20	A1	0010
50	22	75	4.80	2.25	1.40	14.00	7.8	46.8	0.27	A2	0011
50	47	75	2.30	1.50	1.20	12.00	11.4	68.4	0.32	A3	0012
75	6.8	100	15.60	3.70	3.00	63.00	6.5	39.0	0.18	A1	0013
75	10	100	10.60	3.00	2.40	50.40	7.9	47.4	0.22	A2	0014
75	15	100	7.10	2.00	1.60	33.60	9.7	58.2	0.32	A3	0015
100	4.7	150	19.80	3.75	3.00	63.00	7.2	43.2	0.18	A1	0016
100	6.8	150	13.70	3.00	2.40	50.40	8.7	52.2	0.22	A2	0017
100	10	150	9.30	2.00	1.60	33.60	10.5	63.0	0.32	A3	0018

TABLE II. Style CUR01 capacitors.

DC rated voltage at 85°C & 105°C	Nominal Capacitance	DC surge voltage at 85°C & 105°C	Equivalent series resistance at 25°C (max)		Maximum impedance at 25°C (100 kHz)	Maximum impedance at -55°C (100 kHz)	Maximum DC leakage		Maximum AC ripple current at 85°C (20-100 kHz)	Case size	DASH NUMBER 1/
			120 Hz	10 kHz			25°C	105°C			
Volts	µF	Volts	Ohms		Ohms	Ohms	µA		Amperes		
10	470	15	0.37	0.16	0.12	1.80	7.2	43.2	1.17	B1	0101-
10	680	15	0.25	0.13	0.10	1.50	8.7	52.2	1.44	B2	0102-
15	330	20	0.52	0.18	0.12	1.80	9.1	54.6	1.17	B1	0103-
15	470	20	0.37	0.14	0.11	1.65	10.8	64.8	1.44	B2	0104-
15	680	20	0.25	0.12	0.10	1.50	13.0	78.0	1.58	C1	0105-
15	1000	20	0.17	0.08	0.06	0.90	15.8	94.8	2.25	C2	0106-
30	150	40	0.71	0.33	0.18	1.80	12.2	73.2	0.81	B1	0107-
30	220	40	0.48	0.22	0.17	1.70	14.8	88.8	1.35	C1	0108-
30	330	40	0.32	0.20	0.10	1.00	18.2	109.2	1.80	C2	0109-
50	68	75	1.60	0.60	0.40	4.00	13.7	82.2	0.58	B1	0110-
50	100	75	1.10	0.40	0.28	2.80	16.7	100.2	0.90	B2	0111-
50	150	75	0.71	0.32	0.25	2.50	20.4	122.4	1.08	C1	0112-
50	220	75	0.48	0.20	0.17	1.70	24.7	148.2	1.44	C2	0113-
75	47	100	2.30	0.60	0.40	8.40	17.1	102.6	0.58	B1	0114-
75	68	100	1.60	0.40	0.35	7.35	20.6	123.6	0.72	C1	0115-
75	100	100	1.10	0.30	0.22	4.62	25.0	150.0	0.90	C2	0116-
100	22	150	4.30	0.83	0.55	11.55	15.6	93.6	0.58	B1	0117-
100	33	150	2.80	0.42	0.35	7.35	19.1	114.6	0.86	C1	0118-
100	68	150	1.40	0.30	0.20	4.20	27.5	165.0	1.12	C2	0119-

1/ Complete part number will include a letter symbol to indicate failure rate level (i.e., 0101M, 0112R, etc.)

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case:
Type - Tubular.
Material - Metal.

Terminals: See figure 1.

DC rated voltage: See tables I and II.

Reverse voltage: 1.5 volts maximum.

Capacitance (Cap.) value: See tables I and II.

Cap. tolerance: -10 +100%.

Failure rate level (CUR01 only): M, P, R, and S.

Burn-in (CUR01 only): In accordance with MIL-C-39018.

Operating temperature range: -55°C to 105°C.

DC leakage (DCL): See tables I and II.

Equivalent series resistance (ESR): See tables I and II.

Low temperature exposure:
DCL - See tables I and II.
Cap. - See tables I and II.
ESR - See tables I and II.

Terminal strength: Method 211, MIL-STD-202, test conditions A (3 pounds) and D (3 rotations).

Stability at reduced and high temperatures:
Step 1 (at +25°C):
DCL - See tables I and II.
Cap. - See tables I and II.
ESR - See tables I and II.
Step 2 (at -55°C):
Impedance - See tables I and II.
ΔCap. - Shall not exceed the following percentages of the step 1 value:
7 Vdc: ±35 percent.
10 to 30 Vdc, inclusive: ±25 percent.
50 to 100 Vdc, inclusive: ±20 percent.
Step 3 (at +25°C):
DCL - See tables I and II.
ΔCap. - Shall not exceed ±5 percent of step 1 value.
ESR - See tables I and II.
Step 4 (at +105°C):
DCL - See tables I and II.
ΔCap. - Shall not exceed ±30 percent of step 1 value.
ESR - See tables I and II.
Step 5 (at +25°C):
DCL - See tables I and II.
ΔCap. - Shall not exceed ±5 percent of step 1 value.
ESR - See tables I and II.

Life: Method 108, MIL-STD-202.
For qualification: Condition F (2,000 hours) at 85°C.
DCL - See tables I and II.
ΔCap. - Shall not exceed ±10 percent of initial measured value.
ESR - Not more than 130 percent of initial requirement.

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For quality conformance: Performance check, condition B(250 hours) at 85°C (non-ER).
DCL - See table I.
ΔCap. - Shall not exceed ±10 percent of initial measured value.
ESR - See table I.
For quality conformance: Continuation test (1,750 hours) at 85°C (non-ER).
DCL - See table I.
ΔCap. - Shall not exceed ±15 percent of initial measured value.
ESR - Not more than 150 percent of initial requirement.
For quality conformance: 10,000 hours (ER) at 85°C.
DCL - At 25°C, initial limit; at 85°C, 125 percent of initial limit.
ΔCap. - Shall not exceed ±25 percent of initial measured value.
ESR - Not more than 200 percent of initial requirement.

Vibration:
Low frequency - Method 201, MIL-STD-202.

Thermal shock and immersion:
Thermal shock: Method 107, MIL-STD-202, condition A.
Immersion: Method 104, MIL-STD-202, condition B (2 cycles, 30 minutes).
DCL - See tables I and II.
ΔCap. - Shall not exceed ±5 percent of initial measured value.
ESR - See tables I and II.
Case insulation: Shall meet initial requirements.

Surge voltage:
DCL - See tables I and II.
ΔCap. - Shall not exceed ±6 percent of initial measured value.
ESR - See tables I and II.

Moisture resistance: Method 106, MIL-STD-202.
DCL - See tables I and II.
ΔCap. - Shall not exceed ±6 percent of initial measured value.
ESR - See tables I and II.
Case insulation: Shall meet initial requirements.

High temperature exposure (500 hours at 105°C):
DCL - Not more than 200 percent of initial requirement.
ΔCap. - Shall not exceed ±10 percent of initial measured value.
ESR - Not more than 115 percent of initial requirement.

Penetration of solvents: In accordance with MIL-C-39018.

High temperature verification: In accordance with MIL-C-39018 at 105°C with rated voltage applied.
DCL - See tables I and II.
ΔCap. - Shall not exceed ±10 percent of initial measured value.
ESR - Not more than 150 percent of initial requirement.

Reverse voltage aging: In accordance with MIL-C-39018.

Marking: In accordance with MIL-C-39018.

Part number: M39018/08-(dash number from table I or II).

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APPLICATION NOTES:

1. Ripple current - Capacitors will withstand the rms ripple current at 20 kHz through 100 kHz as given in table I or II.

The maximum ripple current values apply with the following qualifications:

The sum of the applied dc voltage plus the peak value of the impressed ac voltage must not exceed the dc working voltage of the capacitor.

The sum of the applied dc voltage plus the peak value of the impressed ac voltage must be such that not more than 1.5 volt reversal is experienced by the capacitor.

If capacitors are operated at a frequency less than 20 kHz, the rated 20 kHz rms ripple current must be multiplied by the factors in table III.

TABLE III. Factors for determining ripple current at various frequencies.

Rated voltage	Frequency (in Hertz)				
	60	120	400	1K - 19K	20 K - 100K
0-75	.60	.71	.75	.82	1.0
76-100	.43	.54	.71	.79	1.0

Where capacitors are operated at a temperature other than +85°C, the permissible rms ripple current must be multiplied by the factors in table IV.

TABLE IV. Factors for determining ripple current at various temperatures.

Ambient temperature	Factor
+105°C	0.5
+ 85°C	1.0
+ 65°C	1.4
+ 45°C	1.7

When capacitors are operated at both a temperature and frequency other than 85°C and less than 20 kHz, both the multiplying factors described in tables III and IV must be used to obtain the permissible rms ripple current.

Under no conditions of temperature or frequency, regardless of the multipliers given above, should the rms ripple current exceed 5 amperes.

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2. Insulation and grounding - These capacitors possess an indeterminate insulation resistance between the cathode terminal and the case. Where a potential difference exists between the cathode terminal and the chassis or other metallic mounting surfaces (which are usually at ground potential), the capacitors should be mounted in such a manner as to insulate the container from such mounting surfaces.

3. Operating life (CUR01 only) - These capacitors are expected to have a useful operating life in excess of ten years when subjected to typical circuit environments and operating conditions. Life can be extended by derating with regard to envirmental and electrical requirements.

4. Shelf life - The general shelf life expectancy of these capacitors is in excess of ten years provided the storage temperatures do not exceed 40°C. Storage of these capacitors at higher temperatures may decrease their shelf life (see MIL-STD-1131).

5. High altitude - These capacitors are capable of operating at altitudes up to 200,000 feet (60,690 meters).

NOTE: Revision letters are not used to denote changes due to extensiveness of changes.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Preparing activity:
Navy - EC

(Project 5910-1420-1)

Review activities:
Navy - SH
Air Force - 17, 85, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 19

Agent:
DLA - ES

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